BAA1027

Data Analytics:

Machine Learning and Advanced Python

Martinas Jucys Brady

22330516

Word Count: 2,561 words

Date: 19/04/2026

DCU BUSINESS SCHOOL

DCU

Abstract

Brain tumours are among the most life-threatening diseases, and their accurate classification is essential for effective and timely treatment planning. This study presents a machine learning framework for the automated classification of brain tumours from MRI scans into the categories of glioma, meningioma, pituitary tumour and no tumour. The approach used in this study flattens raw MRI images into high-dimensional feature vectors, applies PCA for dimensionality reduction while retaining 95% of the explained variance and classifies the reduced features into four machine learning algorithms (SVM, Random Forest, KNN and Naïve Bayes). These classifiers are further combined through an ensemble model. The framework is then evaluated using a Brain Tumor Classification MRI dataset from Kaggle that included 3,264 MRI images. The results show that KNN achieved the highest individual accuracy of 76.4%, followed by a three-model ensemble (comprised of SVM, RF and KNN) at 75.4%. The findings confirm that PCA based dimensionality reduction combined with ensemble classification provides an efficient and viable option for brain tumour classification.

Introduction

Brain tumours are one of the most life-threatening types of diseases and occur through an abnormal growth of cells inside the brain or its surrounding bone and tissue. They can be broadly categorised as benign or malignant and can be graded on a range of I to IV (Pichaivel et al., 2022). It is extremely important to accurately classify brain tumours as it is critical in giving patients the best treatment and increasing the chances of survival.

Magnetic Resonance Imaging (MRI) is the standard diagnosis mechanism used for brain tumours due to the production of high contrast visualisations of soft tissue structures without having to expose patients to ionising radiation (Leung et al., 2014). However, MRI scans are extremely time consuming and prone to error when analysed manually. Diagnostic error rates in radiology departments can reach up to 30%, with perceptual errors accounting for approximately 70% of these mistakes (Zhang et al., 2023). This highlights the need for automated classification systems that can assist radiologists in accurately identifying brain tumour types.

Machine learning has opened up the possibilities for automated medical image analysis, with several studies exploring the roles that deep learning and machine learning can play in this development. Aamir et al. (2022) uses deep learning techniques, while Yildirim et al. (2023) uses a combination of CNN based feature extraction with an SVM classifier to achieve 95.4% accuracy. However it must be noted that deep learning methods like those used in those studies require large datasets and computational resources, which aren’t always available.

This study proposes an alternative approach that relies on classical machine learning techniques. Rather than training a neural network, the framework would extract features by flattening raw MRI images into high dimensional vectors with the application of PCA for dimensionality reduction and the classification of these features using an ensemble of four supervised learning algorithms. This ensemble uses soft voting to combine probability estimates from each individual classifier. This was taken from a scikit-learn algorithm selection flowchart that recommended SVM, KNN and ensemble classifiers for labelled classification tasks with more than 50 samples.

Related Works

The classification of brain tumours from MRI images is heavily covered by existing studies that use both machine learning and deep learning approaches. The following papers are all key inspirations for the processes and techniques used in this study.

Pichaivel et al. (2022) provided an overview of brain tumour types, pathophysiology and different diagnostic methods. It is noted that meningiomas are the most common brain tumours in adults, accounting for 36% of all brain tumours and are benign the majority of the time. Gliomas are the second most common and pituitary tumours are third, accounting for 15% of cases. This paper was particularly useful in understanding why certain tumour types are harder to classify than others. For example, Gliomas can occur across multiple lobes of the brain taking on variable shapes and boundaries, making them extremely difficult to classify accurately. In contrast, meningiomas are typically well-defined, round masses on the brains' surface. Understanding the differences between each is essential in interpreting classification results. The visual similarities between a glioma and meningioma on MRI scans have been identified as a major issue in accurate classification.

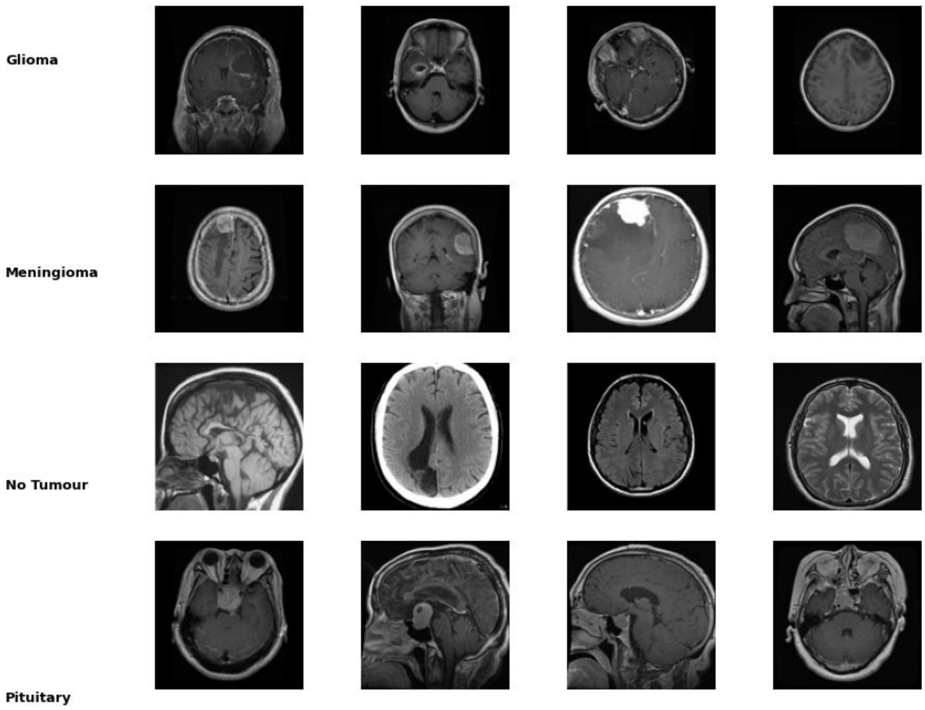

Figure 1: Sample MRI Images by Tumour Type

Zhang et al. (2023) provided an overview of diagnostic errors in radiology departments. This study categorised errors into either perceptual, where abnormalities are present but not noticed, and cognitive, where abnormalities were identified but misinterpreted. It is reported that approximately 75% of malpractice lawsuits against radiologists relate to diagnostic imaging errors. The main factors that lead to missed diagnoses were identified as small lesion sizes, similarities to surrounding tissue and unusual locations. This provides a strong motivation to develop an accurate automated classification system that can support radiologists in providing a grounded second opinion that reduces the likelihood of human error significantly.

Saeedi at al. (2023) analysed both deep learning and machine learning methods for MRI based brain tumour detection. This study compared multiple algorithmic approaches and concluded that combining different algorithms can yield higher performance. This finding directly influenced the decision in adopting an ensemble strategy, where multiple classifiers are used rather than relying on one single algorithm.

Yildirim et al. (2023) developed a model that classified brain tumour types using features extracted from pretrained “EfficientNetB0” and “ShuffleNet” architectures, with the eventual accuracy of 95.4%. This hybrid approach of deep feature extraction followed by the use of a classifier is similar to the methodology used here, except that PCA based feature reduction is used rather than CNN-based feature extraction.

Oh et al. (2024) explored various machine learning algorithms for brain tumour detection and highlighted the importance of feature engineering and efficient model selection. This reinforced the use of multiple classifiers used in this study.

Allahem et al. (2025) used a hybrid model that combined “Vision Transformer-based” feature extraction with PCA for dimensionality reduction and Random Forest for classification of different brain cancer types. Their use of PCA to reduce high dimensional features before applying a classifier directly reflects the approach taken in this study. The paper also shows that using PCA is an effective step in computational cost reduction while preserving the most discriminative information in the data.

Alshomrani (2025) analysed the broad challenges and advances being made in brain tumour classification, covering machine learning, deep learning and hybrid approaches. It was highlighted that ensemble methods are consistently more effective than single-model approaches by reducing both bias and variance. It was also noted that combining feature extraction with classifiers are effective when posed with limited computational resources. This directly supports the model design choices in this study.

Altogether, these studies use both deep learning and machine learning approaches for effective brain tumour classification. However it was also highlighted that deep learning methods such as CNN rely on large datasets and computational resources. This puts forward that using PCA based feature extraction with an ensemble of classifiers allows the production of an efficient alternative.

Methodology

Dataset

The dataset used in this study is the Brain Tumor Classification (MRI) dataset from Kaggle (Bhuvaji et al., 2020). This dataset contains 3,264 contrast enhanced MRI images which are divided into four classes: glioma tumour (826 training and 100 testing images), meningioma tumour (822 training and 115 testing images), pituitary tumour (827 training and 74 testing images), and no tumour (395 training and 105 testing images). The dataset was already split into training and testing folders, 2870 and 394 images respectively. A clear imbalance initially exists, with the no tumour class containing approximately half the number of images of the other classes.

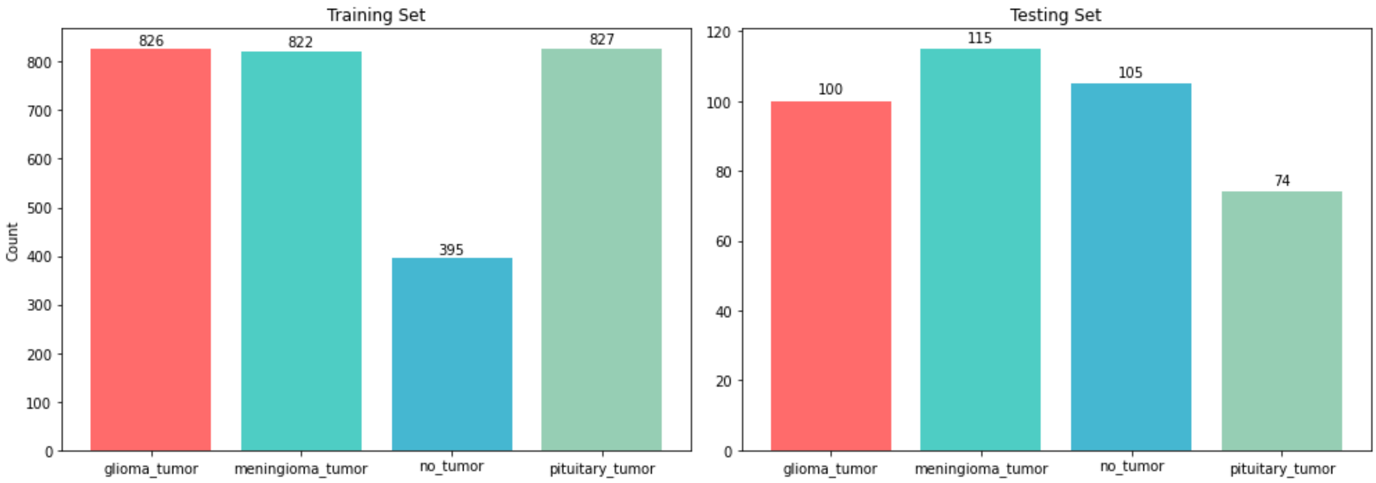

Figure 2: Evident Imbalance with no_tumour data

Preprocessing

All images were resized to 128x128 pixels and converted to a RGB format to ensure uniformity across the dataset as the original images varied in size and resolution. Pixel values were also normalised to [0, 1] by dividing by 255 as this ensures that all features are on the same scale and prevents on feature dominating over others. Text labels, such as ‘glioma_tumour’, were converted to numerical encodings using scikit-learn’s LabelEncoder.

Initial CNN and overfitting issue

Initially, a Convolutional Neural Network (CNN) was built and trained on the dataset. The CNN contained three convolutional blocks with 32, 64 and 128 filters respectively. This was followed by batch normalisation, max pooling and dropout layers. The model also included data augmentation techniques such as random horizontal flipping, rotation and zoom to increase the diversity within the training data.

However, despite taking all these actions, the CNN persistently over the training data. Training accuracy reached values of ~75%, but validation accuracy continuously remained at approximately 29%. The validation loss also increased with each epoch rather than decreasing, which is a clear indicator of overfitting. This all occurred because the dataset of 2,296 images was simply too small to train a CNN from scratch.

This frustrating and time-consuming experience made me switch to a PCA based approach. Rather than attempting to learn spatial features through convolutional layers, the new approach interpreted each image as a flat vector of pixels and through the application of PCA the most important patterns were extracted. While this sacrifices the ability to capture spatial relationships between pixels, it avoids the frustrating overfitting problem and allows for the use of machine learning algorithms that are better suited to smaller datasets.

Feature Extraction and Dimensionality Reduction

Each 128x128 image was flattened into a feature vector of 49,152 dimensions. This captured all the pixel information, but the high dimensionality created a problem for the classifiers used. This issue increases the computational cost, leads to overfitting and causes the 'curse of dimensionality' where distances between data points become less meaningful in high dimensional spaces. This was a particular problem for KNN, which specifically relied on distance measurements between data points.

To fix this issue, "StandardScaler" was then applied to ensure zero mean and unit variance across all features. PCA was then applied to reduce the dimensionality while retaining 95% of the explained variance. This reduced the feature space from

49,152 to 726 key components, which is a 98.5% reduction in dimensionality with the preservation of the most important information (Allahem et al. 2025).

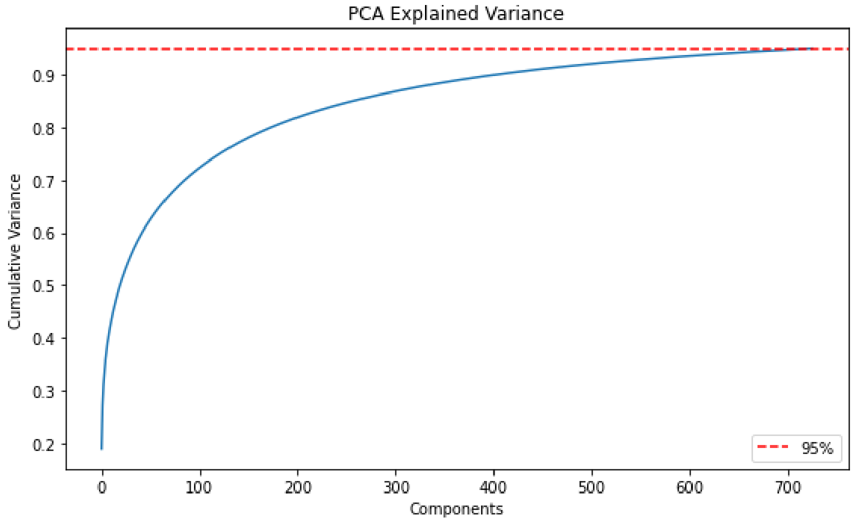

Figure 3: PCA cumulative explained variance plot

<u>Classification Algorithms</u>

SVM, Random Forest, KNN and Naïve Bayes were the four algorithms selected, each representing a different approach to the classification problem. This selection was based on the scikit-learn estimator selection guide, which recommends these classifiers for classification tasks, as well as information learned from the literature review.

Support Vector Machine (SVM) finds the optimal hyperplane that separates classes with maximum margins, using an RBF kernel to handle non-linearly separable data. SVM is well suited for high dimensional spaces and is widely used for brain tumour classification (Yildirim et al., 2023). Random Forest was used to construct 200 decision trees that were each trained on a random subset of features and data, with the final prediction based on majority voting. This bagging approach reduces the overfitting that would be found in a single decision tree (Allahem et al., 2025). K-Nearest Neighbours (KNN) is used to classify new instances based on the seven closest training examples using a distance-weighted voting model. This allowed closer neighbours to have more influence on the prediction.

In contrast, Naïve Bayes was included deliberately as a control classifier that was expected to have a poor performance. This allowed the validation that the other classifiers were learning the patters rather than exploiting any sort of data leakage. Naïve Bayes assumes all features are independent and is a known limitation for image data where neighbouring pixel values are highly correlated.

Ensemble Classification

The four classifiers were combined using scikit-learn's 'VotingClassifier' with soft voting. This approaches allows the strengths of each classifier to be leveraged, where SVM's is the ability to find decision boundaries, Random Forest's robustness and KNN's sensitivity to patters. Two ensembles were evaluated; a four-model ensemble and a three-model ensemble which excluded Naïve Bayes. This was to assess whether a weak classifier degrades the overall ensemble performance.

Evaluation Metrics

Model performance was evaluated using accuracy, precision, recall and F1 score with weighted averaging. Accuracy measures the overall proportion of correct predictions. Precision measures how often positive predictions are actually correct, and recall measures how many actual positive cases are correctly identified. These two measures are extremely important in a medical setting as it could lead to unnecessary stress or treatment and missed diagnosis or delay in treatment respectively. F1 score is the mean of precision and recall, thus providing a single metric. A confusion matrix was also generated for an analysis of the performance of the classifiers for each brain tumour type.

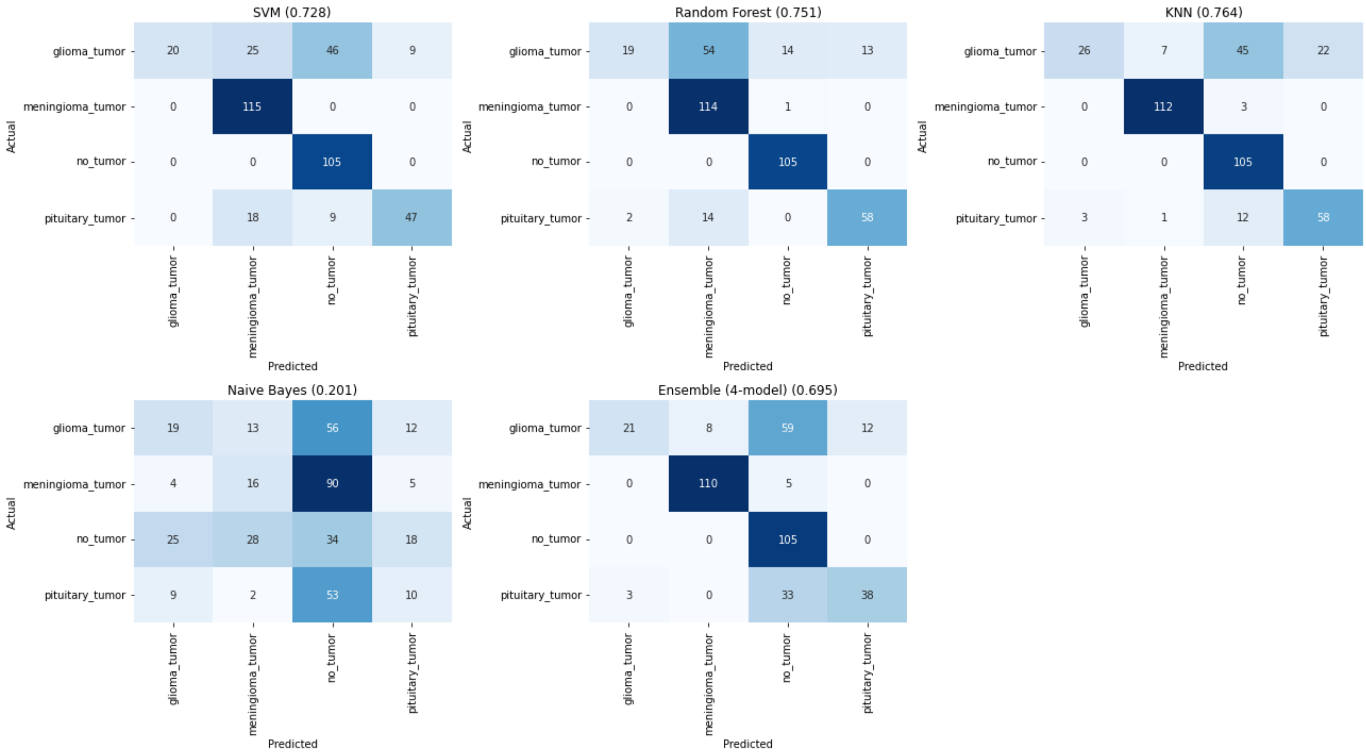

Figure 4: Confusion Matrix for each model

Results

Classification Results

Model	Acc	Prec	Rec	F1
SVM	0.7284	0.7988	0.7284	0.6775
Random Forest	0.7513	0.7991	0.7513	0.7028
KNN	0.7640	0.8057	0.7640	0.7293
Naive Bayes	0.2005	0.2444	0.2005	0.2003
Ensemble (4-model)	0.6954	0.7754	0.6954	0.6590
Ensemble (SVM+RF+KNN)	0.7538	0.7999	0.7538	0.7121

Figure 5: Classification Results for each model

Discussion

These results reveal numerous important findings. KNN achieved the highest individual accuracy at 76.4%, closely followed by Random Forest at 75.1% and SVM at 72.8%. The three-model ensemble, which was composed of SVM, RF and KNN, achieved 75.4%, which was an improvement on the individual values of SVM and Random Forest. This result supports the statement that ensemble methods consistently outperform single model approaches by leveraging complementary classifier strengths (Alshomrani, 2025).

Naïve Bayes performed extremely poorly at 23.9%, which could be considered the same as making a random decision. This is largely due to its assumption that all features are independent of each other. In image data this assumption does not hold because neighbouring pixel values are highly correlated. Its inclusion in this study was deliberate and serves an important purpose as a control classifier. If Naïve Bayes has unexpectedly performed well then it would suggest that there is an issue with the data. This also led to the four-model ensemble performing poorly with a 69.5% accuracy rating. This highlights that ensembles are only effective when the classifiers within contribute to the predictions in a meaningful manner.

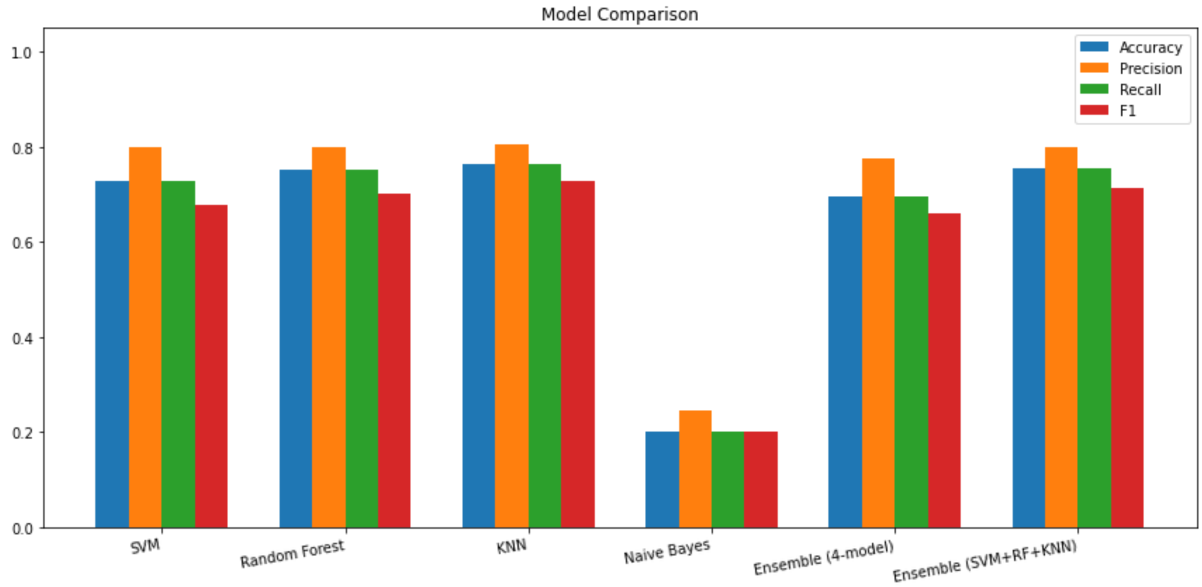

Figure 6: Model Comparison Visualisation

An analysis of values for each tumour type uncovers large variation. Meningioma was classified with 98% recall, while glioma had only 23% . This aligns with observations from Pichaivel et al. (2022), who stated that gliomas exhibit highly variable visual characteristics across different brain regions. The no tumour class also achieved a 100% recall, which indicates that healthy brain scans are distinct from any scans containing any type of tumour. Saeedi et al. (2023) also highlighted the issues surrounding the difficulty of distinguishing gliomas from meningiomas in automated classifications systems, confirming that this is a widely recognised challenge rather than one that only this study faced.

	precision	recall	f1-score	support
glioma_tumor	0.92	0.23	0.37	100
meningioma_tumor	0.86	0.98	0.92	115
no_tumor	0.63	1.00	0.77	105
pituitary_tumor	0.78	0.76	0.77	74
accuracy			0.75	394
macro avg	0.80	0.74	0.71	394
weighted avg	0.80	0.75	0.71	394

Figure 7: Three Model Ensemble Breakdown by Tumour Type

The overall accuracy of 76% for this study compared with higher accuracy values of 95+% associated with deep learning models is expected (Yildirim et al., 2023). CNN based feature extraction captures special patterns that are highly discriminative for image classification, while PCA based models that flatten the pixels lose these special patterns. This trade-off between accuracy and simplicity is a deliberate as it prioritises interpretability.

Limitations

The dataset used in this study is relatively small, with only 3,264 images, and also has a clear imbalance with the no tumour class being underrepresented. The PCA based approach discards the spatial information that is critical in being able to see the difference between similar tumour types. Future work could incorporate texture-based feature extraction methods such as Histogram of Oriented Gradients or move to using pretrained CNNs (Aamir et al., 2022).

This study only classifies four categories of brain abnormalities (glioma, meningioma, pituitary tumour, and no tumour), which represents a small subset of the many different types of brain tumours that exist (Pichaivel et al., 2022). A system used in the healthcare industry would need to account for a much larger variation of tumour types and subtypes to be useful.

Conclusion

This study combined PCA-based dimensionality reduction with an ensemble of supervised learning algorithms in a machine learning framework for brain tumour classification. PCA reduced the 49,152 image features to just 726 components while retaining 95% of the variance. KNN achieved the highest individual accuracy at 76.4%, while the three-model ensemble (composed of SVM, RF and KNN) achieved 75.4%. Naïve Bayes, as expected, proved to be unusable for image classification due to its independence assumption. An analysis of brain tumour types showed that glioma tumours were extremely difficult to classify, which is supported by literature. The findings of this study confirm that PCA combined with ensemble learning is a viable and efficient approach for brain tumour classification, though accuracy remains well below deep learning methods. Future work should explore more complex feature extraction while using a larger dataset.

Bibliography

Aamir, M. *et al.* (2022) ‘A deep learning approach for brain tumor classification using MRI images’, *Computers and Electrical Engineering*, 101, p. 108105. Available at: https://doi.org/10.1016/j.compeleceng.2022.108105.

Allahem, H. *et al.* (2025) ‘A Hybrid Model of Feature Extraction and Dimensionality Reduction Using ViT, PCA, and Random Forest for Multi-Classification of Brain Cancer’, *Diagnostics*, 15(11), p. 1392. Available at: https://doi.org/10.3390/diagnostics15111392.

Alshomrani, F. (2025) ‘Challenges and Advances in Classifying Brain Tumors: An Overview of Machine, Deep Learning, and Hybrid Approaches with Future Perspectives in Medical Imaging’, *Current Medical Imaging*, 21, p. e15734056365191. Available at: https://doi.org/10.2174/0115734056365191250602124819.

Bhuvaji, S. *et al.* (2020) ‘Brain Tumor Classification (MRI)’. Kaggle. Available at: https://doi.org/10.34740/KAGGLE/DSV/1183165.

Leung, D. *et al.* (2014) ‘Role of MRI in Primary Brain Tumor Evaluation’, *Journal of the National Comprehensive Cancer Network*, 12(11), pp. 1561–1568. Available at: https://doi.org/10.6004/jnccn.2014.0156.

Oh, A. *et al.* (2024) ‘Machine learning approach to brain tumor detection and classification’. arXiv. Available at: https://doi.org/10.48550/arXiv.2410.12692.

Patil, S. and Kirange, D. (2023) ‘Ensemble of Deep Learning Models for Brain Tumor Detection’, *Procedia Computer Science*, 218, pp. 2468–2479. Available at: https://doi.org/10.1016/j.procs.2023.01.222.

Pichaivel, M. *et al.* (2022) ‘An Overview of Brain Tumor’, *Brain Tumors*. IntechOpen. Available at: https://doi.org/10.5772/intechopen.100806.

Saeedi, S. *et al.* (2023) ‘MRI-based brain tumor detection using convolutional deep learning methods and chosen machine learning techniques’, *BMC Medical Informatics and Decision Making*, 23, p. 16. Available at: https://doi.org/10.1186/s12911-023-02114-6.

scikit-learn (no date) *13. Choosing the right estimator*, *scikit-learn*. Available at: https://scikit-learn/stable/machine_learning_map.html (Accessed: 26 March 2026).

Toufiq, D.M., Sagheer, A.M. and Veisi, H. (no date) ‘A Review on Brain Tumor Classification in MRI Images’.

Yildirim, M. *et al.* (2023) ‘Detection and classification of glioma, meningioma, pituitary tumor, and normal in brain magnetic resonance imaging using deep learning-based hybrid model’, *Iran Journal of Computer Science*, 6(4), pp. 455–464. Available at: https://doi.org/10.1007/s42044-023-00139-8.

Zhang, L. *et al.* (2023) ‘Diagnostic error and bias in the department of radiology: a pictorial essay’, *Insights into Imaging*, 14, p. 163. Available at: https://doi.org/10.1186/s13244-023-01521-7.

APPENDIX I: STUDENT DECLARATION OF ACADEMIC INTEGRITY

Students may be required to submit work for assessment in a variety of means, for example physical submission or electronic submission as per the lecturer's instructions**. In all cases students must make a declaration of academic integrity, either by physically completing such a declaration and submitting it with their assignment or engaging appropriately with the electronic version of the declaration on Loop.** Assignments submitted such that the form has not been included, or the electronic equivalent has been circumvented, will not be accepted.

DECLARATION

NAME:	Martinas Jucys Brady
STUDENT ID NUMBER	22330516
PROGRAMME	INTB4
MODULE CODE	BAA1027
ASSIGNMENT TITLE	Research Paper
SUBMISSION DATE	19/04/2026

I understand that the University regards academic misconduct as grave and serious.
I have read and understood the DCU Academic Integrity Policy. I accept the penalties that may be imposed should I engage in academic misconduct.
I have identified and included the source of all facts, ideas, opinions and viewpoints of others in the assignment references. Direct quotations, paraphrasing, discussion of ideas from books, journal articles, internet sources, module text, or any other source whatsoever are acknowledged and the sources cited are identified in the assignment references.
I have not made unauthorised use of artificial intelligence aids.
I declare that this material, which I now submit for assessment, is entirely my own work and has not been taken from the work of others save and to the extent that such work has been cited and acknowledged within the text of my work.
I have used the DCU library referencing guidelines (available at https://www.dcu.ie/library/citing-referencing **and/or** the appropriate referencing system recommended in the assignment guidelines and/or programme documentation.
By signing this form or by submitting material online I confirm that this assignment, or any part of it, has not been previously submitted by me or any other person for assessment on this or any other course of study.
By signing this form or by submitting material for assessment online I confirm that I have read and understood the DCU Academic Integrity Policy

Signature: Martinas Jucys Brady

Date: 19/04/2026

APPENDIX II: STUDENT DECLARATION OF GENERATIVE AI

Note: Replace "__" with an "**X**" below to mark/tick against the appropriate options.

__ I acknowledge that I ***have not*** used any Generative AI and LLM models to carry out any task within this assignment.

X I acknowledge that I ***have*** used Generative AI and LLM models to carry out the following tasks as a part of this assignment and I provide the details of such use below.

AI Model Used	**Used For**
Google Gemini	Used to help summarise and digest complex information found in the literature review.
Grammarly	Used as an aid in proofreading and fixing grammatical errors.

Student name: Martinas Jucys Brady

Dated: 19/04/2026